

Vishal R

Bangalore, India | 8147741585 | vishalravi163@gmail.com
linkedin.com/in/vishal-ravi-653a8a33b | github.com/vishal-163 | vishalr.vercel.app

Summary

Aspiring Software Engineer and Full Stack Developer Intern (Computer Science student) with hands-on experience building scalable, production-style web and AI-integrated applications using React.js, Next.js, Node.js, Express.js, and PostgreSQL. Experienced in REST API design, database schema design, authentication, authorization, and LLM API integration (OpenAI, Gemini). Currently architecting a 4-metric IoT-based defence health-monitoring system. Strong foundation in system design, Agile development practices, and version control.

Technical Skills

Languages: JavaScript, TypeScript, Python, Java, SQL, HTML5, CSS3

Frontend: React.js, Next.js 14, Tailwind CSS, shadcn/ui, Framer Motion, Responsive Design

Backend: Node.js, Express.js, REST API Design, JWT Authentication, Role-Based Access Control (RBAC), Flask

Database: PostgreSQL, Supabase, MySQL, Database Schema Design, Row-Level Security

AI, APIs: OpenAI API, Gemini API, Prompt Engineering, LLM Integration

Tools, Practices: Git, GitHub, Docker, Vercel, Netlify, VS Code, Agile, Scrum, Continuous Integration, Continuous Deployment, System Design

Projects

AI Trip Planner

Completed

Flutter, Supabase, OpenAI API, Gemini API, PostgreSQL

- Built a cross-platform mobile application that generates a complete personalized travel itinerary, including destinations, budget breakdown, and day-by-day schedule, from a single user input form.
- Designed a 3-tier modular architecture (Flutter frontend, Supabase backend, dedicated AI service layer) enabling independent scaling and seamless switching between 2 LLM providers, OpenAI and Gemini.
- Engineered deterministic, schema-validated JSON output from LLM responses, eliminating malformed-response failures in UI rendering.
- Implemented secure authentication (email + OAuth) with row-level security and a normalized relational schema for efficient, consistent data handling.
- Built full CRUD functionality with real-time sync via Supabase subscriptions, enabling live multi-device itinerary updates.

Smart Military Vest — IoT Defence System

In Progress

ESP32, Sensors, LoRa, GSM, GPS

- Designing a defence-grade wearable system for real-time soldier health monitoring across 4 vital metrics: heart rate, SpO2, temperature, and motion.
- Built multi-sensor integration across 2 communication protocols (I2C and SPI) with on-device preprocessing to reduce raw data volume transmitted over the network.
- Architecting dual-channel communication (LoRa + GSM/4G) with AES-256 encryption for redundant, secure data transmission in low-connectivity zones.
- Developing an automated distress-signal alert system triggered by configurable health-threshold breaches.
- Planning a scalable backend architecture using MQTT and a time-series database to support a live command-centre analytics dashboard.

Education

Bachelor of Engineering in Computer Science and Engineering

Expected 2027

K.S. Institute of Technology, Bangalore

Relevant Coursework: Data Structures and Algorithms, Database Management Systems, Operating Systems, Computer Networks, Object-Oriented Programming, System Design

Languages

English, Kannada, Hindi, Telugu (Native), Tamil (Fluent)